

Literate Emacs Loading

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1 Package Initialization

This initializes the servers to install packages from. I use Elpa, Melpa, Melpa Development, and Org Elpa.

```
(require 'package)
(add-to-list 'package-archives '("elpa" . "https://elpa.gnu.org/packages/"))
(add-to-list 'package-archives '("melpa" . "https://stable.melpa.org/packages/"))
(add-to-list 'package-archives '("melpa-devel" . "https://melpa.org/packages/"))
(add-to-list 'package-archives '("org" . "https://orgmode.org/elpa/"))

(package-initialize)
```

2 Environment and Appearance

2.1 Appearance

This customizes the appearance of the raw emacs look. It modifies the fore/back ground, the text, fonts, and comments. It also disables the startup and toolbar modes.

```
(setq debug-on-error t) ;; Debug on Errors
(setq image-auto-resize nil) ;; Resize Images

(setq-default indent-tabs-mode nil) ;; Set Default Tab to Two Spaces
(setq-default tab-width 2)

(add-to-list 'default-frame-alist '(foreground-color . "#FFFFFF")) ;; Background
(add-to-list 'default-frame-alist '(background-color . "#000000")) ;; Foreground
(add-to-list 'default-frame-alist '(fullscreen . maximized)) ;; Start Maximized

(set-frame-parameter nil 'alpha-background 100)
(add-to-list 'default-frame-alist '(alpha-background . 100))

(menu-bar-mode -1)
(scroll-bar-mode -1)
(tool-bar-mode -1)

(set-face-attribute 'default nil :font "Noto Sans Mono")
(set-fontset-font t 'unicode (font-spec :family "Noto Color Emoji"))
(set-face-foreground 'font-lock-comment-face "#C28aa1")
```

```
(require 'org)
(unless (package-installed-p 'org-modern)
  (package-install 'org-modern))
(require 'org-modern)
(add-hook 'org-mode-hook #'org-modern-mode)

(setq org-hide-emphasis-markers t)
(setq org-confirm-babel-evaluate nil)
(setq org-hide-keywords t)
```

```
(custom-set-faces
 '(org-block
  (
   (t
    (:background "#0a0017" :foreground "white" :weight normal)
   )))
```

```
(require 'ansi-color)

(defun org-babel-strip-ansi ()
  (when (org-in-src-block-p)
    (ansi-color-apply-on-region (point-min) (point-max))))

(add-hook 'org-babel-after-execute-hook #'org-babel-strip-ansi)
```

2.2 Environment

```
(setq inhibit-startup-screen t)
(setq backup-directory-alist `(("." . , "~/.emacs.d/.backups")))
(setq make-backup-files t)
(setq backup-by-copying t)
(setq version-control t)
(setq delete-old-versions t)
(setq delete-by-moving-to-trash t)
(setq kept-old-versions 5)
(setq kept-new-versions 10)
(setq auto-save-default t)
(setq auto-save-timeout 20)
(setq auto-save-interval 200)
(defalias 'yes-or-no-p 'y-or-n-p)

(global-auto-revert-mode 1) ;; Refresh Buffers on Change
(setq global-auto-revert-non-file-buffers t)
(setq auto-revert-verbose nil)

(add-to-list 'display-buffer-alist '("\\\\*Buffer List\\\\*"
  ↪ (display-buffer-same-window)))

;; Open Ansi-Term as Default Terminal Emulator
(defun open-ansi-term-here (dir) "Open an ansi-term in DIR."
  (let (
    (default-directory
     (file-name-as-directory
      (expand-file-name dir))))
    (ansi-term (getenv "SHELL"))))

;; Reload the Init File
(defun reload-init ()
  "Reloading Emacs Init"
  (interactive)
  (load-file user-init-file))

(defun ensure-package-and-require (pkg)
```

```

"Install PKG if it isn't installed, then require it."
(unless (package-installed-p pkg)
  (package-install pkg))
(require pkg)

(add-hook 'emacs-startup-hook (lambda () (interactive) (ansi-term (getenv
↪ "SHELL")))))

```

```

(with-eval-after-load 'org
(setq org-preview-latex-process-alist
  '((dvipng :programs ("latex" "dvipng")
    :description "dvi > png"
    :message "you need to install the programs: latex and dvipng."
    :use-xcolor t
    :image-input-type "dvi"
    :image-output-type "png"
    :image-size-adjust (1.0 . 1.0)
    :latex-compiler ("latex -interaction nonstopmode -output-directory
↪ %o %f")
    :image-converter ("dvipng -D %D -T tight -o %O %f")))))

(setq org-latex-listings 'minted)
(setq org-latex-packages-alist '((" " "minted"))))
(setq org-latex-minted-options
  '(("breaklines" "true")
    ("fontsize" "\\scriptsize")
    ("frame" "single"))))

```

```

(add-hook 'org-mode-hook #'hs-minor-mode)
(add-hook 'org-mode-hook #'yas-minor-mode)

```

```

(require 'ob)
(require 'ob-python)
(unless (package-installed-p 'ess)
  (package-install 'ess))
(require 'ess)
(require 'ess-site)
(require 'ob-R)
(require 'ob-julia)

```

```

(org-babel-do-load-languages
 'org-babel-load-languages
 '((emacs-lisp . t)
   (C . t)
   (R . t)
   (python . t)
   (latex . t)
   (shell . t)
   (julia . t)))

```

```

(unless (package-installed-p 'exec-path-from-shell)
  (package-install 'exec-path-from-shell))
(require 'exec-path-from-shell)
(exec-path-from-shell-initialize)

```

```
(require 'tramp)
(with-eval-after-load 'tramp
  (add-to-list 'tramp-remote-path 'tramp-own-remote-path))
```

3 Shortcuts

The remote shortcuts are untracked since they have sensitive information. This file needs to be copied manually.

```
(add-to-list 'load-path "~/.emacs.d/lisp/")
(require 'remote-shortcuts)
(require 'tramp-paths)
```

```
(global-set-key (kbd "C-h C-r") 'reload-init)
(global-set-key (kbd "M-s M-s")
  (lambda ()
    (interactive)
    (ansi-term (getenv "SHELL"))))
```

```
(add-hook 'c++-mode-hook
  (lambda ()
    (local-set-key (kbd "C-c C-v") 'compile)))
```

Enable the snippet shortcuts everywhere

```
(unless (package-installed-p 'yasnipet)
  (package-install 'yasnipet))
(require 'yasnipet)
(yas-global-mode 1)
```

4 Plugins

4.1 Magit

```
(unless (package-installed-p 'magit)
  (package-install 'magit))
(require 'magit)
(setq magit-diff-refine-hunk 'all)
(global-auto-revert-mode 1)
```

4.2 Treemacs

```
;; Prevent Treemacs from trying to load magit-bookmark
(with-eval-after-load 'treemacs-bookmarks
  (define-minor-mode treemacs--dummy-magit-bookmark-mode
```

```

    "Dummy mode to satisfy Treemacs bookmark requirement."))

(provide 'magit-bookmark) ;; pretend the feature exists

```

```

(unless (package-installed-p 'treemacs)
  (package-install 'treemacs))
(unless (package-installed-p 's)
  (package-install 's))

(require 'treemacs)
(require 's)

(with-eval-after-load 'treemacs
  (defun treemacs-custom-filter (file _)
    (or (s-ends-with? ".aux" file)
        (s-ends-with? ".log" file)
        (s-ends-with? ".o" file)
        (s-ends-with? ".pyc" file)
        (s-ends-with? ".bak" file)
        (s-ends-with? "~" file)))

  (add-to-list 'treemacs-ignored-file-predicates
    #'treemacs-custom-filter))

```

4.3 Projectile

```

(unless (package-installed-p 'projectile)
  (package-install 'projectile))

(require 'projectile)

```

5 Programming

5.1 General

```

(add-hook 'prog-mode-hook #'hs-minor-mode)

```

Flymake is not needed since I am using Eglot and LSP servers.

```

(setq flymake-diagnostic-functions
  (remove 'flymake-cc flymake-diagnostic-functions))
(add-hook 'c-mode-hook
  (lambda ()
    (setq-local flymake-diagnostic-functions nil)))
(add-hook 'c++-mode-hook
  (lambda ()
    (setq-local flymake-diagnostic-functions nil)))
(add-hook 'python-mode-hook
  (lambda ()
    (setq-local flymake-diagnostic-functions nil)))

```

```
(unless (package-installed-p 'imenu-list)
  (package-install 'imenu-list))
(require 'imenu-list)
(setq imenu-list-focus-after-activation t
      imenu-list-auto-resize t
      imenu-list-size 30)
(global-set-key (kbd "C-c i") #'imenu-list-smart-toggle)
(add-hook 'prog-mode-hook #'outline-minor-mode)
```

5.2 Python

This activates the Emacs communication with Pyenv Virtualenv. In the local directories, it is able to automatically detect the proper pyrightconfig.json file, and use the server installed in the corresponding environment.

```
(unless (package-installed-p 'pyenv-mode)
  (package-install 'pyenv-mode))
(require 'pyenv-mode)
(pyenv-mode 1)

(defun pyenv-version-path ()
  "Return the path to the python executable in the current pyenv/virtualenv."
  (interactive)
  (let* ((v-name (if (and (boundp 'pyenv-mode-version) pyenv-mode-version)
                     pyenv-mode-version
                     (string-trim (shell-command-to-string "pyenv version-name")))))
    (path (condition-case nil
            (let ((root (string-trim
                        (shell-command-to-string
                          (concat "pyenv prefix " v-name)))))
              (concat root "/bin/python"))
            (error "python3"))))
    (when (called-interactively-p 'interactive)
      (message "Resolved Python: %s" path))
    path))
```

```
(unless (package-installed-p 'ein)
  (package-install 'ein))

(require 'ein)
(setq ein:jupyter-server-command
      "~/pyenv/versions/jupyter/bin/jupyter")
(add-to-list 'auto-mode-alist '("\\.ipynb\\'" . ein:notebook-mode))
```

```
(setq ein:jupyter-default-server-command
      (lambda ()
        (expand-file-name "bin/jupyter" pyenv-virtual-env)))
(add-to-list 'auto-mode-alist '("\\.ipynb\\'" . ein:notebook-mode))
```

5.3 C/C++

```
(unless (package-installed-p 'dap-mode)
  (package-install 'dap-mode))
(require 'dap-gdb-lldb)
(require 'dap-cpptools)
(require 'dap-ui)
(dap-ui-mode 1)
;; Optional: automatically enable DAP for C/C++ buffers
(add-hook 'c++-mode-hook #'dap-mode)
(add-hook 'c-mode-hook #'dap-mode)
```

5.4 Julia

```
(unless (package-installed-p 'julia-mode)
  (package-install 'julia-mode))
(require 'julia-mode)
```

5.5 R

```
(with-eval-after-load 'ob-R
  (setq org-babel-R-command
    "R --slave --no-save --no-restore -e
    ↪ \"options(repos=c(CRAN='https://cloud.r-project.org'))\""
  )

  (with-eval-after-load 'ess-r-mode
    (setq ess-r-command
      "R --slave --no-save --no-restore -e
      ↪ \"options(repos=c(CRAN='https://cloud.r-project.org'))\""
    )
  )
)
```

5.6 Eglot

Not necessarily needed, but just to ensure portability

```
(require 'eglot)
(setq eglot-events-buffer-size 10000000)
(setq eglot-report-progress t)
```

```
(defun eglot-python-contact (<_>)
  "Find the pyright-langserver.
If remote, return the remote-local path (without TRAMP prefix).
If local, return the absolute path."
  (let* ((python-path (pyenv-version-path))
         (bin-dir (file-name-directory python-path))
         (pyright-bin (expand-file-name "pyright-langserver" bin-dir)))
    ;; We use file-local-name to strip the "/ssh:user@host:" prefix
    ;; if it exists, because TRAMP process-start needs the remote-side path.
    (list (file-local-name pyright-bin) "--stdio")))

(with-eval-after-load 'eglot
  (setq-default eglot-server-programs
```



```
(assq-delete-all 'python-mode eglot-server-programs))
(add-to-list 'eglot-server-programs `(python-mode . ,#'eglot-python-contact)))

(add-hook 'python-mode-hook #'eglot-ensure)
```

```
(add-hook 'c-mode-hook #'eglot-ensure)
(add-hook 'c++-mode-hook #'eglot-ensure)

(with-eval-after-load 'eglot
  (add-to-list 'eglot-server-programs
    '(c++-mode
      . ("clangd")))
  (add-to-list 'eglot-server-programs
    '(c-mode
      . ("clangd")))
)
```

```
(add-hook 'julia-mode-hook #'eglot-ensure)
(with-eval-after-load 'eglot
  (add-to-list 'eglot-server-programs
    '(julia-mode .
      ("julia" "--startup-file=no" "--history-file=no"
       "-e"
       "using LanguageServer, SymbolServer;
       depot_path = get(ENV, \"JULIA_DEPOT_PATH\", \"\");
       project_path = Base.current_project();
       server = LanguageServer.LanguageServerInstance(
         stdin, stdout, project_path, depot_path);
       run(server);"))))
)
```

5.7 Corfu

```
(unless (package-installed-p 'corfu)
  (package-install 'corfu))
(require 'corfu)
(require 'corfu-popupinfo)

(global-corfu-mode 1)
(setq corfu-auto t)
(setq corfu-cycle t)
(setq corfu-quit-no-match 'separator)
(setq corfu-auto-delay 0)
(setq corfu-auto-prefix 1)

(define-key corfu-map (kbd "TAB") 'corfu-next)
(define-key corfu-map (kbd "<tab>") 'corfu-next)
(define-key corfu-map (kbd "S-TAB") 'corfu-next)
```

5.8 Eldoc

```
(unless (package-installed-p 'eldoc-box)
  (package-install 'eldoc-box))
```

```
(require 'eldoc-box)
(add-hook 'eglot-managed-mode-hook #'eldoc-box-hover-mode)

(setq eldoc-documentation-strategy 'eldoc-documentation-compose)
(setq eldoc-box-max-pixel-width 500)
(setq eldoc-box-max-pixel-height 400)
```

6 Conclusion

```
(provide 'init)
```